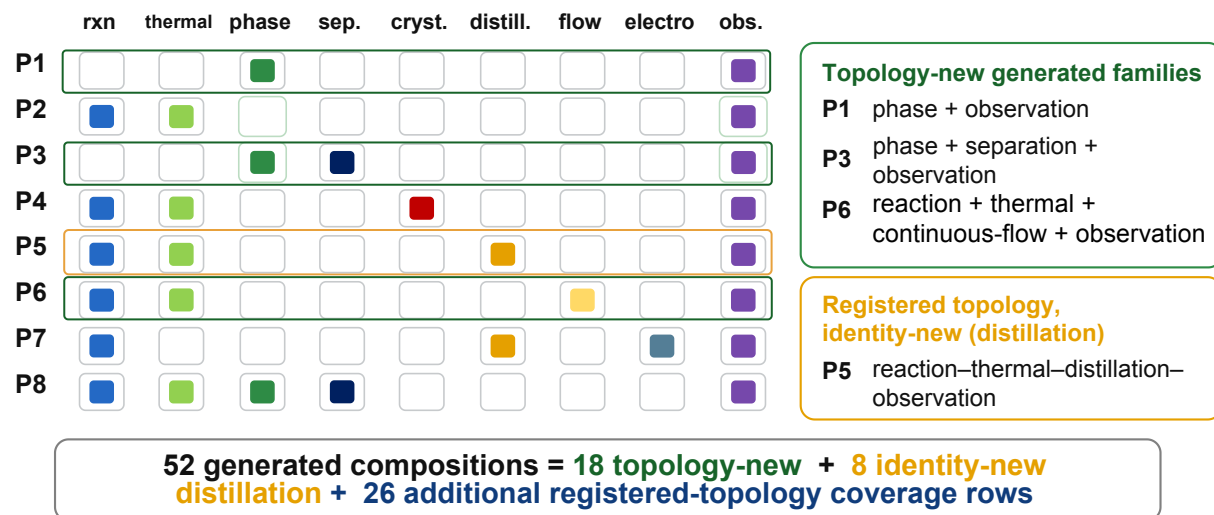
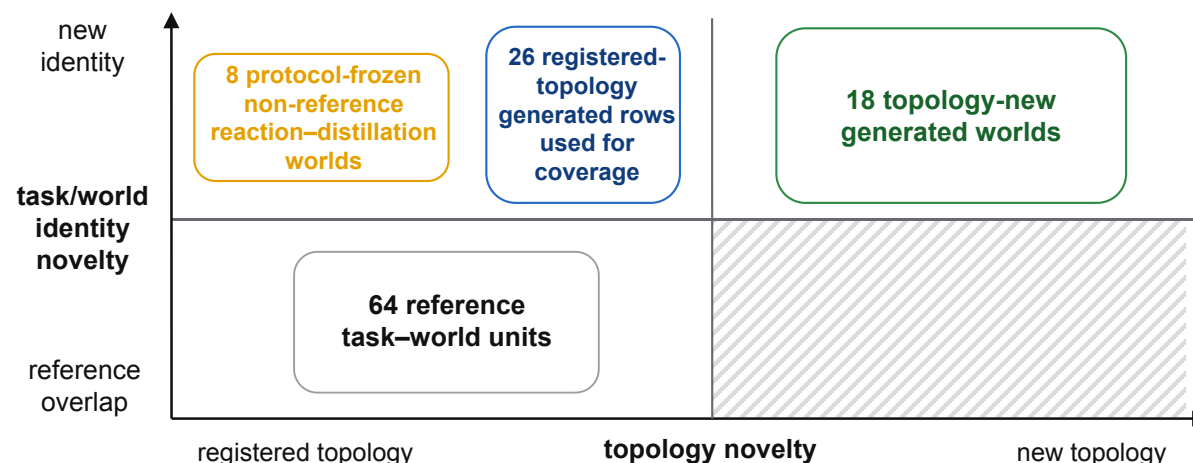


A. From reference tasks to generated world families



B. Topology novelty vs. world-identity novelty



C. What actually varies in generated worlds

1 reaction – thermal – observation	heat (K)	350	●●●●●	390
	duration (s)	600	●●●●●	1,800
2 reaction – thermal – distillation – observation	reaction (K / s)	350 / 600	●●●●●	390 / 1800
	evaporation (K / s)	325 / 300	●●●●●	345 / 900
	distillation (K / s)	350 / 900	●●●●●	390 / 2400
	reflux	1.0	●●●●●	3.0
	transfer	0.65	●●●●●	0.95
3 reaction – thermal – flow – observation	flow (mL min ⁻¹)	0.5	●●●●●	5.0
	residence (s)	60	●●●●●	600
	temperature (K)	330	●●●●●	390
4 reaction – electrochemistry – observation	potential (V)	0.5	●●●●●	1.8
	current (mA)	25	●●●●●	150
	electrolysis (s)	300	●●●●●	1,800

D. Composition changes the experiment surface

phase–observation		phase–separation–observation	
Workflow	add solvent → add reagent → measure → terminate	Workflow	mix → settle → separate → wash → transfer
Public instruments	pH, UV–vis, final assay	Public instruments	HPLC, final assay
reaction–thermal–flow–observation		reaction–thermal–distillation–observation	
Workflow	set flow → set temperature → run → sample	Workflow	heat → quench → evaporate → distill → collect
Public instruments	HPLC, GC, final assay	Public instruments	HPLC, GC, final assay

Composition changes topology, operating conditions, workflow order, and public instrument surface under one common contract.